Capstone Project note – 1

Banking – Probability of Default

Data Science and Business Analytics

Great Learning

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Contents

Content	Page Number
1. Introduction of the business problem	3
1a. Defining problem statement	3
1b. Need of the study/project	3
1c. Understanding business/social opportunity	4
2. Data Report	4
2a. Understanding how data was collected in terms of time, frequency, and methodology	4
2b. Visual inspection of data (rows, columns, descriptive details)	5 – 7
2c. Understanding of attributes (variable info, renaming if required)	8 – 10
3. Exploratory data analysis	10
3a. Univariate analysis (distribution and spread for every continuous attribute, distribution of data in categories for categorical ones)	10 - 12
3b. Bivariate analysis (relationship between different variables, correlations)	12 -15
3c. Removal of unwanted variables (if applicable)	16
3d. Missing Value treatment (if applicable)	16-18
3e. Outlier treatment (if required)	18 – 19
3f. Variable transformation (if applicable)	20
3g. Addition of new variables (if required)	20
4. Business insights from EDA	21
4a. Is the data unbalanced? If so, what can be done? Please explain in the context of the business	21
4b. Any business insights using clustering (if applicable)	21
4c. Any other business insights	21

List of Figures	Page No
Figure 1: Histogram plots for numerical variables	10
Figure 2: Histogram plot for distribution of has_paid variable	11
Figure 3: Histogram plot for distribution of merchant group variable	11
Figure 4: Histogram plot for distribution of merchant category variable	12
Figure 5: Correlation heatmap for all variables	13
Figure 6: Pair plot for continuous attributes	14
Figure 7: Boxplots for all variables vs default with outliers	15
Figure 8: Boxplots after treating the outliers – 1	18
Figure 9: Boxplots after treating the outliers – 2	19
Figure 10: Boxplots after treating the outliers – 3	19

1. Introduction of the business problem

1a. Defining problem statement

Sol: The problem statement is to build a supervised learning classification model for a credit card company, focusing on predicting the probability of default. The objective is to predict whether a customer will pay their credit card bill or default based on the provided variables in the dataset and predictive modelling task is very important in understanding the riskiness of customers and analysing the potential credit exposure in case of default.

Default prediction models play an integral role in institutions that deal with finances especially those lending money since they help in analysing clients' creditworthiness. By leveraging various account-related, purchase, and delinquency information, these models enable organizations to manage risk effectively and make informed decisions regarding credit allocation.

The aim is to develop a robust predictive classification model that enhances the company's ability to identify and mitigate potential default risks, thereby safeguarding its financial health and ensuring prudent lending practices.

1b. Need of the study/project

Sol: The need for study/project is essential because predictive modeling plays a significant role in the operations of financial institutions, especially those in the lending sector. It helps the client to understand and manage default risks within the credit card banking industry. By accurately predicting the likelihood of customers defaulting on their credit card payments, the company can be protected from potential financial losses further helps in maintaining the financial stability of the company.

1. Understanding Riskiness: The Predictive modeling enables us to explore the behaviours and patterns of customers, which helps us to assess the level of riskiness. This understanding is essential for identifying potential defaulters and managing risk effectively from the short term and long-term financial losses.

2. Analyzing Credit Exposure: By predicting the probability of default, we can gain insights into the potential credit exposure faced by the company. This analysis is crucial for assessing the financial impact of defaulting customers and strategizing risk mitigation measures.

3. Enhancing Decision-making: Default prediction models serve as powerful tools for making informed decisions regarding credit allocation. By leveraging various account-related, purchase, and delinquency information, these models enable organizations to allocate credit resources wisely, maximizing profitability while minimizing risk.

4. Promoting Financial Health: A robust predictive classification model not only safeguards the financial health of the company but also promotes responsible lending practices. By accurately identifying and mitigating potential default risks, the company can uphold its reputation and credibility in the financial market.

In summary, the study/project addresses a critical need within the financial sector, offering a proactive approach to risk management and ensuring the sustainability and growth of the company in the long run.

1c. Understanding business/social opportunity

Sol: In the banking industry, building a complex prediction classification model is crucial for efficiently monitoring bad loan risks and below are the mentioned points is why it is crucial:

1. Enhanced Risk Management: By accurately assessing the chances of credit card users defaulting, banks can better control financial risks. This allows them to use resources more efficiently and prevent unexpected losses.

2. Improved Credit Assessment: These models provide banks with more precise insights into their clients' repayment capabilities, enabling better decisions when offering credit products. This helps in customer segmentation based on risk and better solutions accordingly.

3. Efficient Resource Allocation: Banks can allocate more loans or credit by setting higher risk limits, deploying capital and liquidity where default risk is lower. This strategy saves costs and maximizes profitability.

4.Customer Protection: Identifying loan risks through technology allows banks to implement measures to prevent defaults. This includes providing debt management advice and assistance to struggling customers.

5.Market Competitiveness: Developing advanced predictive modeling capabilities enhances a bank's competitiveness. By presenting themselves as proactive risk managers and offering better options to clients, banks can attract and retain customers.

In summary, creating a predictive classification model to detect default risks offers banking institutions significant opportunities to enhance risk management, improve lending assessments, allocate resources efficiently, and offer better products and services to customers.

2. Data Report

2a. Understanding how data was collected in terms of time, frequency, and methodology

Sol: In this project, the data was collected from a credit card company's database. The dataset contains information about various aspects of credit card accounts, including purchase behaviours, delinquency status, account activity, and demographic details of the customers.

Time and Frequency of Data Collection:

- The data spans over a period of time, covering multiple intervals from the past to the present date.
- It includes variables such as "acct_amt_added_12_24m," representing the total amount of purchases made within specific time frames.
- Other variables like "acct_days_in_dc_12_24m" and "acct_days_in_rem_12_24m" capture the duration of account statuses like debt collection and reminders within defined time periods.

Methodology:

- Data collection methodology involves extracting information from the credit card company's database.
- Variables such as "num_active_inv" and "sum_paid_inv_0_12m" provide insights into the number of active invoices and total paid invoices, respectively.
- The dataset also includes categorical variables like "merchant_category" and "merchant_group," which categorize the merchants associated with the transactions.

- Additionally, demographic details like "age" and "name_in_email" provide further context about the customers.

2b. Visual inspection of data (rows, columns, descriptive details)

Sol: For the visual inspection of data, we need to examine the dataset provided for our credit card default prediction problem. Let us start by loading the dataset and exploring its structure, including the rows, columns, and descriptive details of each column.

- Shape of the dataset: **(Rows - 99979, Columns - 36)**
- Name of the columns in the dataset:

```
Column names of the dataset:
Index(['userid', 'default', 'acct_amt_added_12_24m', 'acct_days_in_dc_12_24m',
       'acct_days_in_rem_12_24m', 'acct_days_in_term_12_24m',
       'acct_incoming_debt_vs_paid_0_24m', 'acct_status',
       'acct_worst_status_0_3m', 'acct_worst_status_12_24m',
       'acct_worst_status_3_6m', 'acct_worst_status_6_12m', 'age',
       'avg_payment_span_0_12m', 'avg_payment_span_0_3m', 'merchant_category',
       'merchant_group', 'has_paid', 'max_paid_inv_0_12m',
       'max_paid_inv_0_24m', 'name_in_email',
       'num_active_div_by_paid_inv_0_12m', 'num_active_inv',
       'num_arch_dc_0_12m', 'num_arch_dc_12_24m', 'num_arch_ok_0_12m',
       'num_arch_ok_12_24m', 'num_arch_rem_0_12m',
       'status_max_archived_0_6_months', 'status_max_archived_0_12_months',
       'status_max_archived_0_24_months', 'recovery_debt',
       'sum_capital_paid_acct_0_12m', 'sum_capital_paid_acct_12_24m',
       'sum_paid_inv_0_12m', 'time_hours'],
      dtype='object')
```

- Description about the dataset:

	count	mean	std	min	25%	50%	75%	max
userid	99977.0	2.998947e+06	1.154211e+06	0.0	2.000260e+06	2.998815e+06	4.000633e+06	4.999868e+06
default	89977.0	1.254543e-01	3.333776e+01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+04
acct_amt_added_12_24m	99977.0	1.225503e+04	3.548133e+04	0.0	0.000000e+00	0.000000e+00	4.937000e+03	1.128775e+06
acct_days_in_dc_12_24m	88141.0	3.573252e-01	4.028733e+01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.183600e+04
acct_days_in_rem_12_24m	88141.0	5.178850e+00	4.594340e+01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.183600e+04
acct_days_in_term_12_24m	88141.0	4.211774e-01	3.997377e+01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.183600e+04
acct_incoming_debt_vs_paid_0_24m	40662.0	2.789992e+00	2.953340e+02	0.0	0.000000e+00	1.520899e-01	6.629926e-01	5.931500e+04
acct_status	45604.0	2.234431e+00	2.546089e+02	1.0	1.000000e+00	1.000000e+00	1.000000e+00	5.437300e+04
acct_worst_status_0_3m	45604.0	2.365165e+00	2.546086e+02	1.0	1.000000e+00	1.000000e+00	1.000000e+00	5.437300e+04
acct_worst_status_12_24m	33216.0	3.347212e+00	3.663034e+02	1.0	1.000000e+00	1.000000e+00	2.000000e+00	6.676100e+04
acct_worst_status_3_6m	42275.0	2.550183e+00	2.806343e+02	1.0	1.000000e+00	1.000000e+00	1.000000e+00	5.770200e+04
acct_worst_status_6_12m	39627.0	2.776062e+00	3.031610e+02	1.0	1.000000e+00	1.000000e+00	1.000000e+00	6.035000e+04
age	99977.0	3.601592e+01	1.300174e+01	0.0	2.500000e+01	3.400000e+01	4.500000e+01	1.000000e+02
avg_payment_span_0_12m	76141.0	1.828429e+01	8.725376e+01	0.0	1.080000e+01	1.490909e+01	2.100000e+01	2.383600e+04
avg_payment_span_0_3m	50672.0	1.596251e+01	2.192071e+02	0.0	8.400000e+00	1.300000e+01	1.828571e+01	4.930500e+04
has_paid	88943.0	8.657455e-01	3.409275e-01	0.0	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
max_paid_inv_0_12m	88943.0	9.362712e+03	1.367242e+04	0.0	2.387500e+03	6.170000e+03	1.140000e+04	2.790000e+05
max_paid_inv_0_24m	88943.0	1.141961e+04	1.543171e+04	0.0	3.685000e+03	7.720000e+03	1.386500e+04	5.385000e+05

num_active_div_by_paid_inv_0_12m	70052.0	3.818166e-01	7.137492e+01	0.0	0.000000e+00	0.000000e+00	1.000000e-01	1.889100e+04
num_active_inv	88943.0	6.259852e-01	1.610529e+00	0.0	0.000000e+00	0.000000e+00	1.000000e+00	4.700000e+01
num_arch_dc_0_12m	88943.0	6.261313e-02	3.821109e-01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.700000e+01
num_arch_dc_12_24m	88943.0	5.922894e-02	3.623208e-01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.300000e+01
num_arch_ok_0_12m	88943.0	7.786448e+00	1.674280e+01	0.0	0.000000e+00	3.000000e+00	8.000000e+00	2.610000e+02
num_arch_ok_12_24m	88943.0	6.846576e+00	1.606794e+01	0.0	0.000000e+00	2.000000e+00	7.000000e+00	3.130000e+02
num_arch_rem_0_12m	88943.0	4.839954e-01	1.395611e+00	0.0	0.000000e+00	0.000000e+00	0.000000e+00	4.200000e+01
status_max_archived_0_6_months	88943.0	8.217398e-01	7.166438e-01	0.0	0.000000e+00	1.000000e+00	1.000000e+00	3.000000e+00
status_max_archived_0_12_months	88943.0	1.074182e+00	7.763897e-01	0.0	1.000000e+00	1.000000e+00	2.000000e+00	5.000000e+00
status_max_archived_0_24_months	88943.0	1.248193e+00	8.205178e-01	0.0	1.000000e+00	1.000000e+00	2.000000e+00	5.000000e+00
recovery_debt	88943.0	3.601925e+00	1.162108e+02	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.641100e+04
sum_capital_paid_acct_0_12m	88943.0	1.086026e+04	2.663062e+04	0.0	0.000000e+00	0.000000e+00	8.959500e+03	5.714750e+05
sum_capital_paid_acct_12_24m	88943.0	6.614946e+03	1.924381e+04	0.0	0.000000e+00	0.000000e+00	1.025000e+02	3.418590e+05
sum_paid_inv_0_12m	88943.0	4.103591e+04	9.459642e+04	0.0	3.395500e+03	1.705700e+04	4.573900e+04	2.962870e+06
time_hours	88943.0	1.534165e+01	5.030877e+00	0.0	1.163181e+01	1.580833e+01	1.955417e+01	2.399972e+01

- Head of the given dataset:

	0	1	2	3	4
userid	4567129.0	2635118.0	4804232.0	1442693.0	4575322.0
default	0.0	0.0	0.0	0.0	0.0
acct_amt_added_12_24m	0.0	0.0	0.0	0.0	0.0
acct_days_in_dc_12_24m	0.0	0.0	0.0	NaN	0.0
acct_days_in_rem_12_24m	0.0	0.0	0.0	NaN	0.0
acct_days_in_term_12_24m	0.0	0.0	0.0	NaN	0.0
acct_incoming_debt_vs_paid_0_24m	0.0	NaN	NaN	NaN	NaN
acct_status	1.0	1.0	NaN	NaN	NaN
acct_worst_status_0_3m	1.0	1.0	NaN	NaN	NaN
acct_worst_status_12_24m	NaN	1.0	NaN	NaN	NaN
acct_worst_status_3_6m	1.0	1.0	NaN	NaN	NaN
acct_worst_status_6_12m	NaN	1.0	NaN	NaN	NaN
age	20.0	50.0	22.0	36.0	25.0
avg_payment_span_0_12m	12.692308	25.833333	20.0	4.6875	13.0
avg_payment_span_0_3m	8.333333	25.0	18.0	4.888889	13.0
merchant_category	Dietary supplements	Books & Magazines	Diversified entertainment	Diversified entertainment	Electronic equipment & Related accessories
merchant_group	Health & Beauty	Entertainment	Entertainment	Entertainment	Electronics
has_paid	1.0	1.0	1.0	1.0	1.0
max_paid_inv_0_12m	31638.0	13749.0	29890.0	40040.0	7100.0
max_paid_inv_0_24m	31638.0	13749.0	29890.0	40040.0	7100.0
name_in_email	no_match	F+L	L1+F	F1+L	F+L
num_active_div_by_paid_inv_0_12m	0.153846	0.0	0.071429	0.03125	0.0
num_active_inv	2.0	0.0	1.0	1.0	0.0
num_arch_dc_0_12m	0.0	0.0	0.0	0.0	0.0
num_arch_dc_12_24m	0.0	0.0	0.0	0.0	0.0
num_arch_ok_0_12m	13.0	9.0	11.0	31.0	1.0
num_arch_ok_12_24m	14.0	19.0	0.0	21.0	0.0
num_arch_rem_0_12m	0.0	3.0	3.0	0.0	0.0
status_max_archived_0_6_months	1.0	1.0	1.0	1.0	1.0
status_max_archived_0_12_months	1.0	2.0	2.0	1.0	1.0
status_max_archived_0_24_months	1.0	2.0	2.0	1.0	1.0
recovery_debt	0.0	0.0	0.0	0.0	0.0

sum_capital_paid_acct_0_12m	0.0	0.0	0.0	0.0	0.0
sum_capital_paid_acct_12_24m	0.0	0.0	0.0	0.0	0.0
sum_paid_inv_0_12m	178839.0	49014.0	124839.0	324676.0	7100.0
time_hours	9.653333	13.181389	11.561944	15.751111	12.698611

- Tail of the given dataset:

	99974	99975	99976	99977	99978
userid	4648093.0	1247657.0	NaN	NaN	0.0
default	NaN	NaN	NaN	NaN	10000.0
acct_amt_added_12_24m	56102.0	0.0	NaN	NaN	0.0
acct_days_in_dc_12_24m	0.0	0.0	NaN	NaN	11836.0
acct_days_in_rem_12_24m	0.0	0.0	NaN	NaN	11836.0
acct_days_in_term_12_24m	0.0	0.0	NaN	NaN	11836.0
acct_incoming_debt_vs_paid_0_24m	0.064175	NaN	NaN	NaN	59315.0
acct_status	1.0	1.0	NaN	NaN	54373.0
acct_worst_status_0_3m	2.0	1.0	NaN	NaN	54373.0
acct_worst_status_12_24m	1.0	NaN	NaN	NaN	66761.0
acct_worst_status_3_6m	2.0	1.0	NaN	NaN	57702.0
acct_worst_status_6_12m	2.0	NaN	NaN	NaN	60350.0
age	31.0	41.0	NaN	NaN	0.0
avg_payment_span_0_12m	17.5	34.666667	NaN	NaN	23836.0
avg_payment_span_0_3m	NaN	37.5	NaN	NaN	49305.0
merchant_category	Decoration & Art	Dietary supplements	NaN	NaN	0
merchant_group	Home & Garden	Health & Beauty	NaN	NaN	0
has_paid	1.0	1.0	NaN	NaN	0.0
max_paid_inv_0_12m	15000.0	13246.0	NaN	NaN	0.0
max_paid_inv_0_24m	15000.0	14817.0	NaN	NaN	0.0
name_in_email	Nick	F1+L	NaN	NaN	0
num_active_div_by_paid_inv_0_12m	0.0	0.0	NaN	NaN	18891.0
num_active_inv	0.0	0.0	NaN	NaN	0.0
num_arch_dc_0_12m	0.0	0.0	NaN	NaN	0.0
num_arch_dc_12_24m	0.0	0.0	NaN	NaN	0.0
num_arch_ok_0_12m	2.0	2.0	NaN	NaN	0.0
num_arch_ok_12_24m	1.0	2.0	NaN	NaN	0.0
num_arch_rem_0_12m	0.0	1.0	NaN	NaN	0.0
status_max_archived_0_6_months	1.0	2.0	NaN	NaN	0.0
status_max_archived_0_12_months	1.0	2.0	NaN	NaN	0.0
status_max_archived_0_24_months	1.0	2.0	NaN	NaN	0.0
recovery_debt	0.0	0.0	NaN	NaN	0.0
sum_capital_paid_acct_0_12m	18339.0	0.0	NaN	NaN	0.0
sum_capital_paid_acct_12_24m	56180.0	0.0	NaN	NaN	0.0
sum_paid_inv_0_12m	34785.0	30602.0	NaN	NaN	0.0
time_hours	9.725278	11.585278	NaN	NaN	0.0

2c. Understanding of attributes (variable info, renaming if required)

To understand the attributes of the dataset and perform any renaming, we will follow these steps:

- Information and summary of the provided dataset.

In the below output it has been observed that there are lot of data points missing in the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99979 entries, 0 to 99978
Data columns (total 36 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   userid                            99977 non-null  float64
 1   default                           89977 non-null  float64
 2   acct_amt_added_12_24m             99977 non-null  float64
 3   acct_days_in_dc_12_24m            88141 non-null  float64
 4   acct_days_in_rem_12_24m           88141 non-null  float64
 5   acct_days_in_term_12_24m          88141 non-null  float64
 6   acct_incoming_debt_vs_paid_0_24m  40662 non-null  float64
 7   acct_status                       45604 non-null  float64
 8   acct_worst_status_0_3m            45604 non-null  float64
 9   acct_worst_status_12_24m          33216 non-null  float64
 10  acct_worst_status_3_6m            42275 non-null  float64
 11  acct_worst_status_6_12m           39627 non-null  float64
 12  age                               99977 non-null  float64
 13  avg_payment_span_0_12m            76141 non-null  float64
 14  avg_payment_span_0_3m             50672 non-null  float64
 15  merchant_category                 99977 non-null  object
 16  merchant_group                    99968 non-null  object
 17  has_paid                          88943 non-null  float64
 18  max_paid_inv_0_12m                88943 non-null  float64
 19  max_paid_inv_0_24m                88943 non-null  float64
 20  name_in_email                     88943 non-null  object
 21  num_active_div_by_paid_inv_0_12m  70052 non-null  float64
 22  num_active_inv                    88943 non-null  float64
 23  num_arch_dc_0_12m                 88943 non-null  float64
 24  num_arch_dc_12_24m                88943 non-null  float64
 25  num_arch_ok_0_12m                 88943 non-null  float64
 26  num_arch_ok_12_24m                88943 non-null  float64
 27  num_arch_rem_0_12m                88943 non-null  float64
 28  status_max_archived_0_6_months    88943 non-null  float64
```

```
 29  status_max_archived_0_12_months   88943 non-null  float64
 30  status_max_archived_0_24_months   88943 non-null  float64
 31  recovery_debt                     88943 non-null  float64
 32  sum_capital_paid_acct_0_12m       88943 non-null  float64
 33  sum_capital_paid_acct_12_24m      88943 non-null  float64
 34  sum_paid_inv_0_12m                88943 non-null  float64
 35  time_hours                        88943 non-null  float64
dtypes: float64(33), object(3)
memory usage: 27.5+ MB
```

- Check the data types of each column to ensure they are appropriate.

```
userid                              float64
default                             float64
acct_amt_added_12_24m               float64
acct_days_in_dc_12_24m              float64
acct_days_in_rem_12_24m             float64
acct_days_in_term_12_24m            float64
acct_incoming_debt_vs_paid_0_24m    float64
acct_status                         float64
acct_worst_status_0_3m              float64
acct_worst_status_12_24m            float64
acct_worst_status_3_6m              float64
acct_worst_status_6_12m             float64
age                                 float64
avg_payment_span_0_12m              float64
avg_payment_span_0_3m               float64
merchant_category                    object
merchant_group                       object
has_paid                            float64
max_paid_inv_0_12m                  float64
max_paid_inv_0_24m                  float64
name_in_email                        object
num_active_div_by_paid_inv_0_12m    float64
num_active_inv                      float64
num_arch_dc_0_12m                   float64
num_arch_dc_12_24m                  float64
num_arch_ok_0_12m                   float64
num_arch_ok_12_24m                  float64
num_arch_rem_0_12m                  float64
status_max_archived_0_6_months      float64
status_max_archived_0_12_months     float64
status_max_archived_0_24_months     float64
recovery_debt                       float64
sum_capital_paid_acct_0_12m         float64
sum_capital_paid_acct_12_24m        float64
sum_paid_inv_0_12m                  float64
time_hours                          float64
dtype: object
```

There are 33 variables with float type and 3 variables with object type in the provided dataset.

- Rename the columns based on the problem statement.

```
Index(['userid', 'default', 'total_purchases_12_24m',
       'days_in_debt_collection_12_24m', 'days_in_reminder_12_24m',
       'days_in_termination_12_24m', 'debt_vs_paid_ratio_0_24m',
       'account_status', 'worst_status_0_3m', 'worst_status_12_24m',
       'worst_status_3_6m', 'worst_status_6_12m', 'age',
       'avg_payment_span_0_12m', 'avg_payment_span_0_3m', 'merchant_category',
       'merchant_group', 'has_paid', 'max_paid_inv_0_12m',
       'max_paid_inv_0_24m', 'name_in_email', 'active_div_by_paid_ratio_0_12m',
       'num_active_invoices', 'num_arch_dc_0_12m', 'num_arch_dc_12_24m',
       'num_arch_ok_0_12m', 'num_arch_ok_12_24m', 'num_arch_rem_0_12m',
       'max_archived_status_0_6_months', 'max_archived_status_0_12_months',
       'max_archived_status_0_24_months', 'recovery_debt',
       'sum_capital_paid_acct_0_12m', 'sum_capital_paid_acct_12_24m',
       'sum_paid_inv_0_12m', 'time_hours'],
      dtype='object')
```

3. Exploratory Data Analysis

3a. Univariate analysis (distribution and spread for every continuous attribute, distribution of data in categories for categorical ones)

Sol: Univariate analysis for numerical variables:

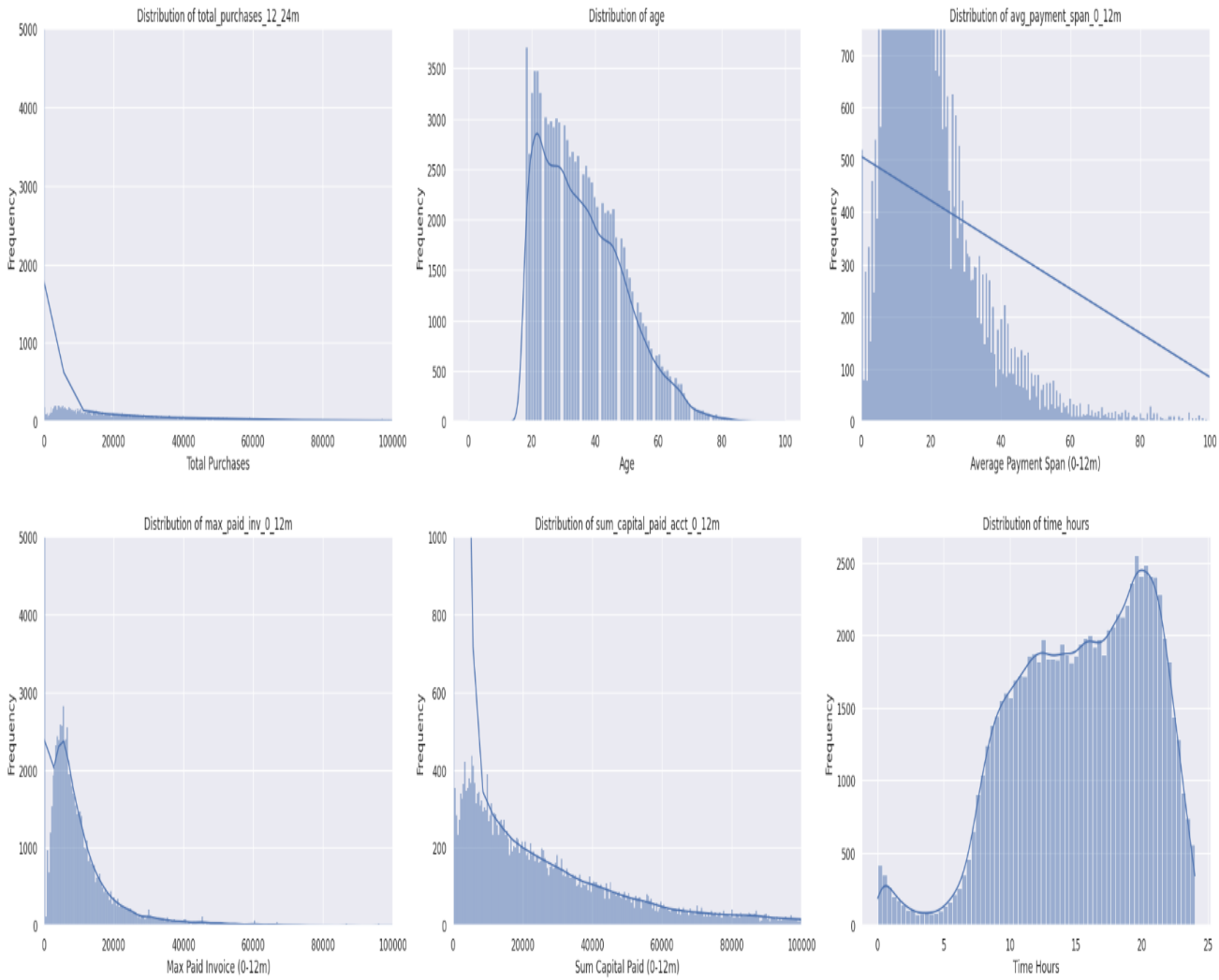

Figure 1: Histogram plots for numerical variables

Univariate analysis for Categorical variables: Has_paid, Merchant group and Merchant category

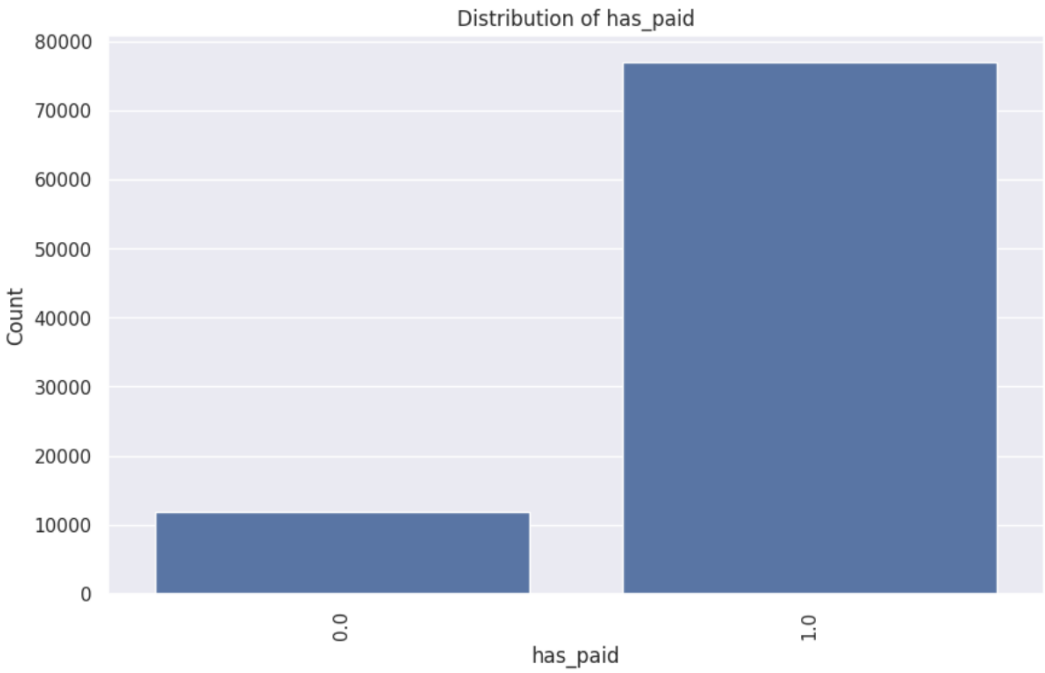

Figure 2: Histogram plot for distribution of has_paid variable

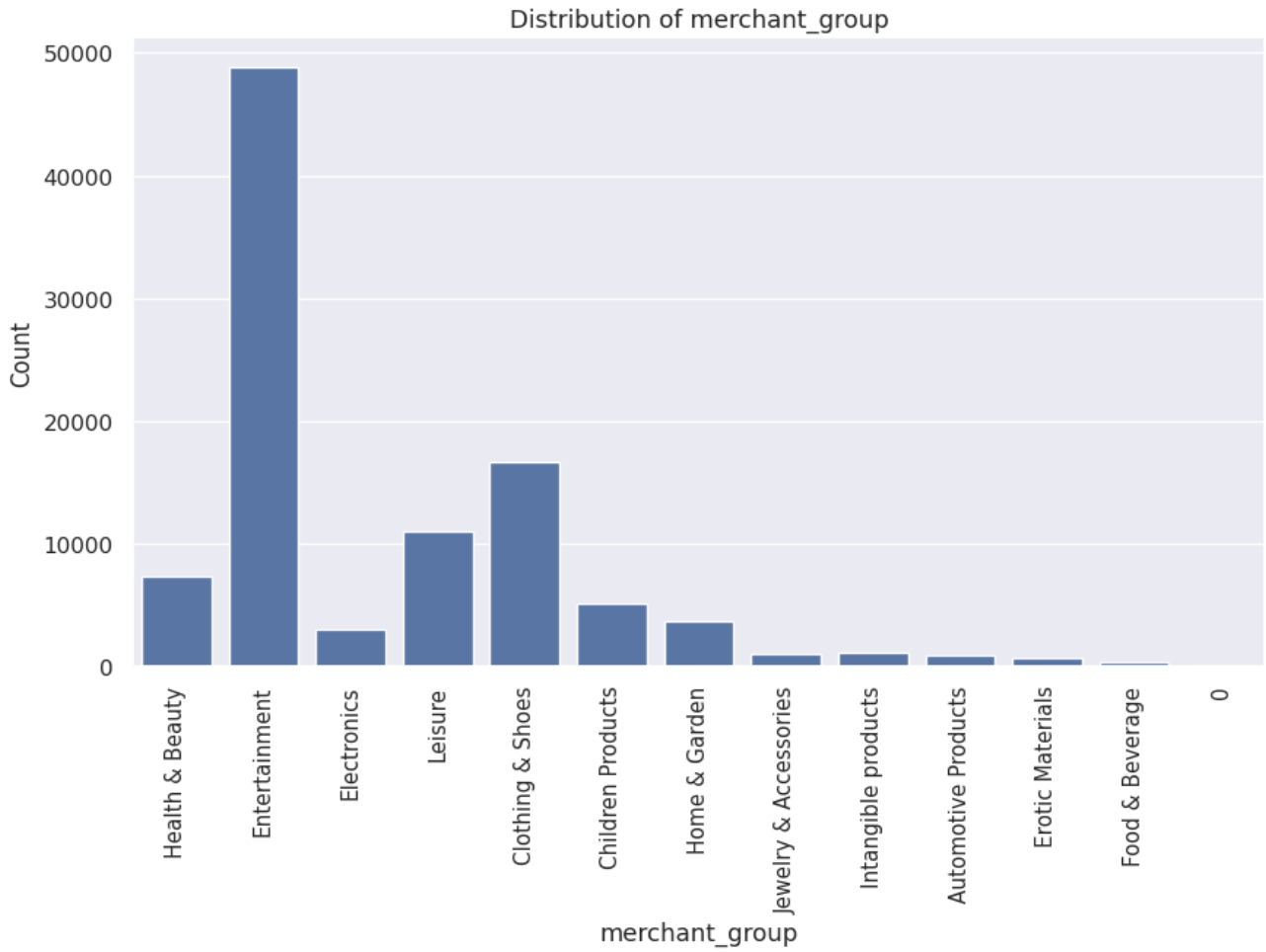

Figure 3: Histogram plot for distribution of merchant group variable

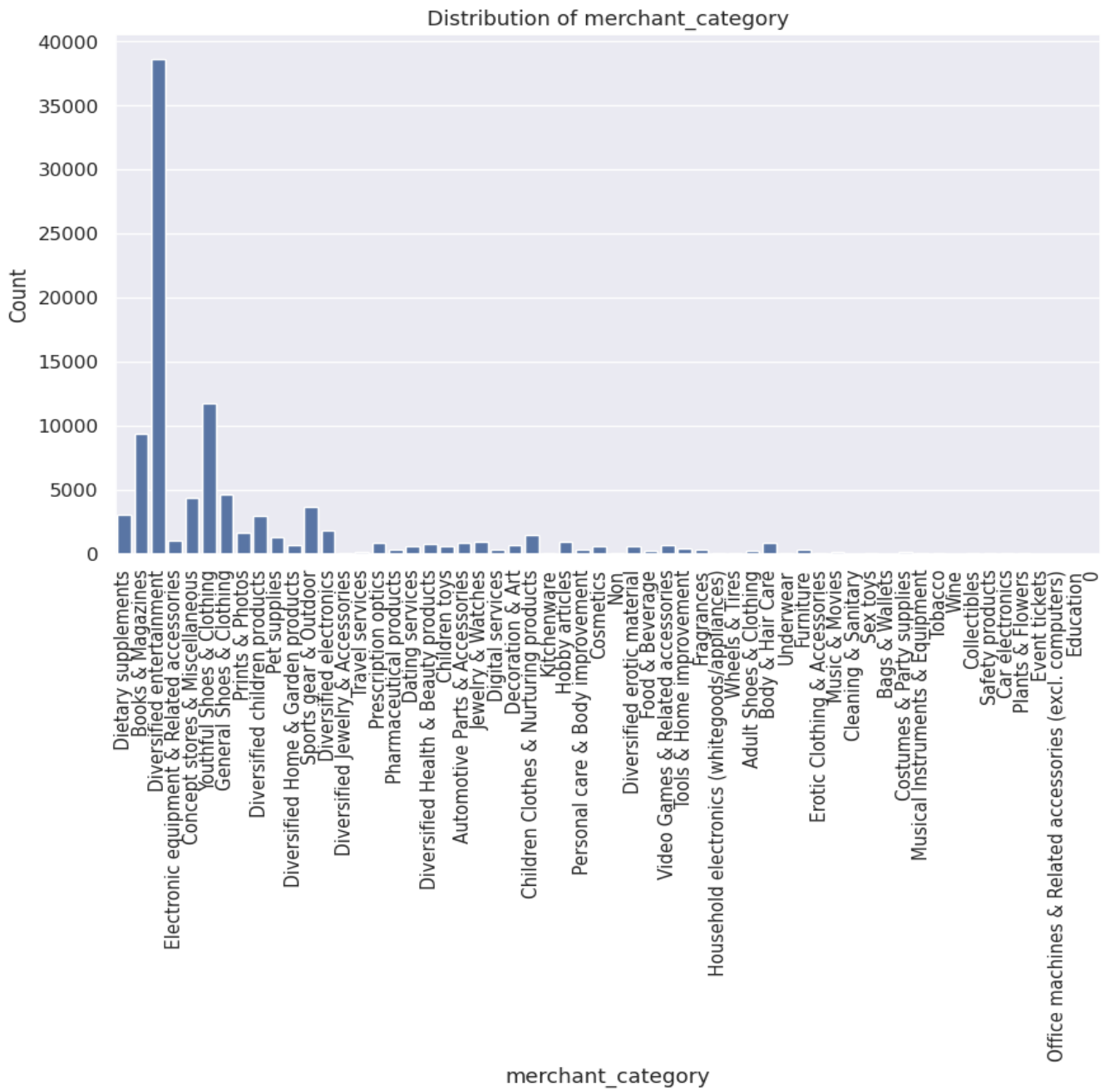

Figure 4: Histogram plot for distribution of merchant category variable

Quick summary of all the above graphs

- Fig1 - Age variable in which the distribution is lies in between 20 – 40 age group and average payment span or day is in between 0 to 40 days.
- Fig3 – Majority in Entertainment followed by clothing & shoes category in the merchant group variable
- Fig4 – Majority of the merchant lies in diversified entertainment

3b. Bivariate analysis (relationship between different variables, correlations)

Correlation matrix for variables:

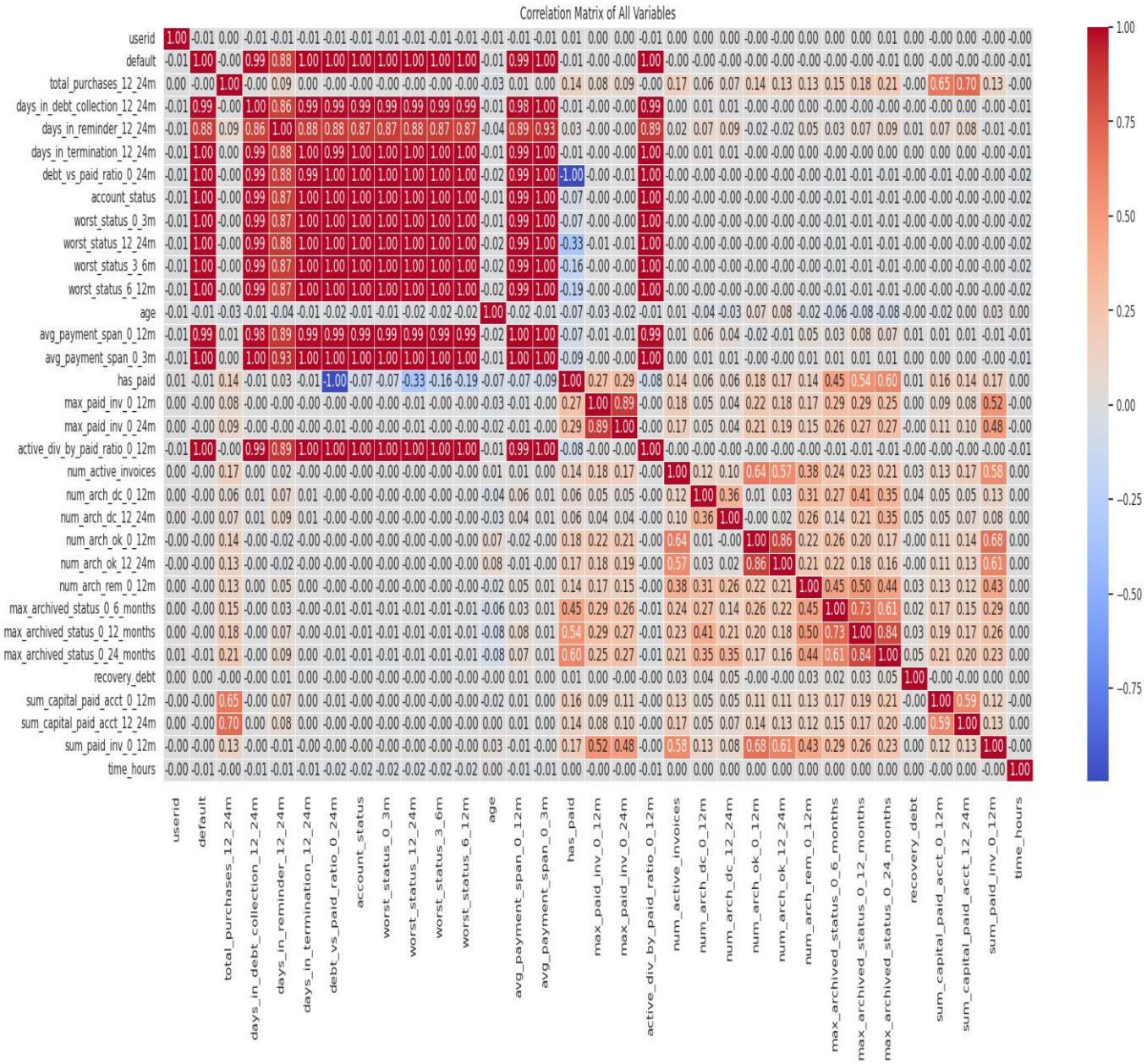

Figure 5: Correlation heatmap for all variables

Pair plots for continuous attributes:

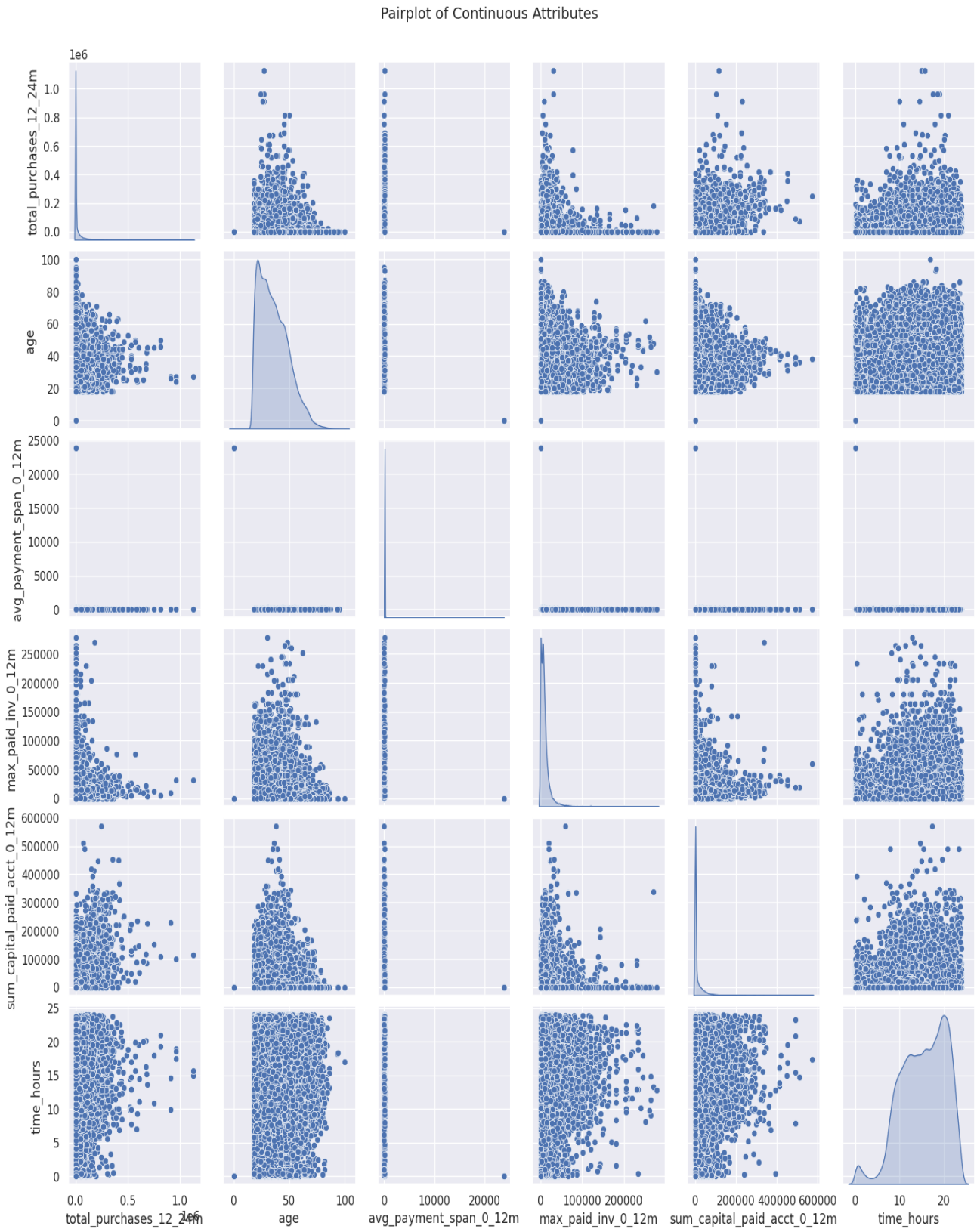

Figure 6: Pair plot for continuous attributes

In the above pair plot it has been observed that there are some relationships with

Bivariate analysis for all the variables with default:

Figure 7: Boxplots for all variables vs default with outliers

3c. Removal of unwanted variables (if applicable)

We found that variable userid which is not required. By applying the drop function, we removed the userid column and the below is the updated output head of the dataset.

	default	total_purchases_12_24m	days_in_debt_collection_12_24m	days_in_reminder_12_24m	days_in_termination_12_24m	debt_vs_paid_ratio_0_24m	account_status
0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
1	0.0	0.0	0.0	0.0	0.0	NaN	1.0
2	0.0	0.0	0.0	0.0	0.0	NaN	NaN
3	0.0	0.0	NaN	NaN	NaN	NaN	NaN
4	0.0	0.0	0.0	0.0	0.0	NaN	NaN

Also, further we will remove the data while treating the missing values

3d. Missing Value treatment (if applicable)

```
Missing Data Count in Ascending Order:
total_purchases_12_24m                  2
merchant_category                       2
age                                     2
merchant_group                         11
default                             10002
num_arch_dc_0_12m                   11036
num_arch_dc_12_24m                  11036
num_arch_ok_0_12m                   11036
num_arch_ok_12_24m                  11036
num_arch_rem_0_12m                  11036
max_archived_status_0_6_months      11036
max_archived_status_0_12_months     11036
max_archived_status_0_24_months     11036
recovery_debt                       11036
sum_capital_paid_acct_0_12m         11036
sum_capital_paid_acct_12_24m        11036
num_active_invoices                 11036
name_in_email                       11036
max_paid_inv_0_12m                  11036
sum_paid_inv_0_12m                  11036
has_paid                            11036
max_paid_inv_0_24m                  11036
time_hours                          11036
days_in_termination_12_24m          11838
days_in_reminder_12_24m             11838
days_in_debt_collection_12_24m      11838
avg_payment_span_0_12m              23838
active_div_by_paid_ratio_0_12m      29927

avg_payment_span_0_3m               49307
worst_status_0_3m                   54375
account_status                      54375
worst_status_3_6m                   57704
debt_vs_paid_ratio_0_24m            59317
worst_status_6_12m                  60352
worst_status_12_24m                 66763
```

```
Percentage of Missing Values for Each Column:
default                            10.004101
total_purchases_12_24m              0.002000
days_in_debt_collection_12_24m     11.840487
days_in_reminder_12_24m            11.840487
days_in_termination_12_24m         11.840487
debt_vs_paid_ratio_0_24m           59.329459
account_status                     54.386421
worst_status_0_3m                  54.386421
worst_status_12_24m                66.777023
worst_status_3_6m                  57.716120
worst_status_6_12m                 60.364677
age                                 0.002000
avg_payment_span_0_12m             23.843007
avg_payment_span_0_3m              49.317357
merchant_category                   0.002000
merchant_group                      0.011002
has_paid                           11.038318
max_paid_inv_0_12m                 11.038318
max_paid_inv_0_24m                 11.038318
name_in_email                      11.038318
active_div_by_paid_ratio_0_12m     29.933286
num_active_invoices                11.038318
num_arch_dc_0_12m                  11.038318
num_arch_dc_12_24m                 11.038318
num_arch_ok_0_12m                  11.038318
num_arch_ok_12_24m                 11.038318
num_arch_rem_0_12m                 11.038318
max_archived_status_0_6_months     11.038318

max_archived_status_0_12_months    11.038318
max_archived_status_0_24_months    11.038318
recovery_debt                      11.038318
sum_capital_paid_acct_0_12m        11.038318
sum_capital_paid_acct_12_24m       11.038318
sum_paid_inv_0_12m                 11.038318
time_hours                         11.038318
dtype: float64
```

Dropping the missing values which is more than 50% data and the updated shape of the dataset is (99979, 29)

Now treating the remaining missing values with mean for all numerical values and mode for categorical values and output is mentioned below

		active_div_by_paid_ratio_0_12m	99979
default	99979	num_active_invoices	99979
total_purchases_12_24m	99979	num_arch_dc_0_12m	99979
days_in_debt_collection_12_24m	99979	num_arch_dc_12_24m	99979
days_in_reminder_12_24m	99979	num_arch_ok_0_12m	99979
days_in_termination_12_24m	99979	num_arch_ok_12_24m	99979
age	99979	num_arch_rem_0_12m	99979
avg_payment_span_0_12m	99979	max_archived_status_0_6_months	99979
avg_payment_span_0_3m	99979	max_archived_status_0_12_months	99979
merchant_category	99979	max_archived_status_0_24_months	99979
merchant_group	99979	recovery_debt	99979
has_paid	99979	sum_capital_paid_acct_0_12m	99979
max_paid_inv_0_12m	99979	sum_capital_paid_acct_12_24m	99979
max_paid_inv_0_24m	99979	sum_paid_inv_0_12m	99979
name_in_email	99979	time_hours	99979

3e. Outlier treatment (if required)

Now by using IQR technique treating outliers in all variables and the output is mentioned below.

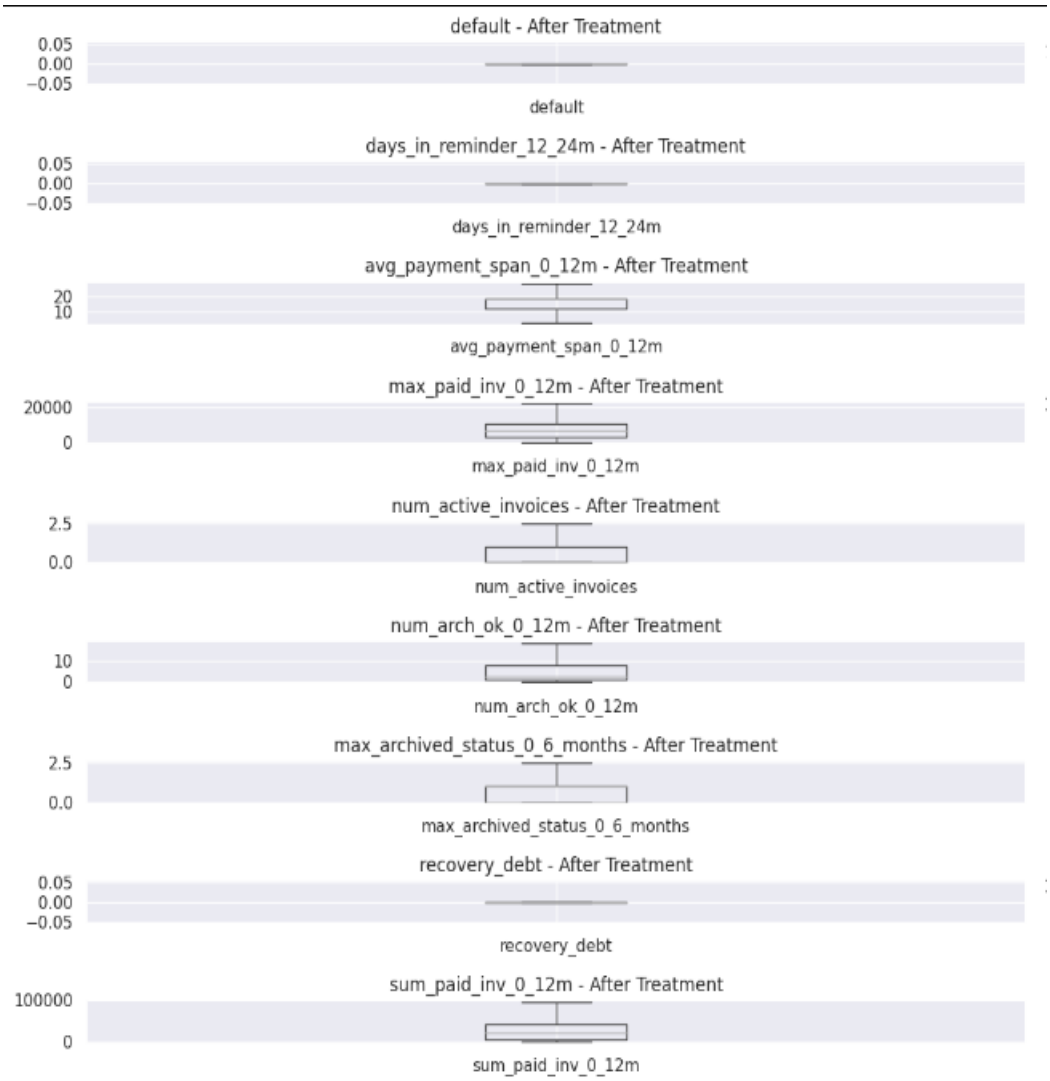

Figure 8: Boxplots after treating the outliers – 1

Figure 9: Boxplots after treating the outliers – 2

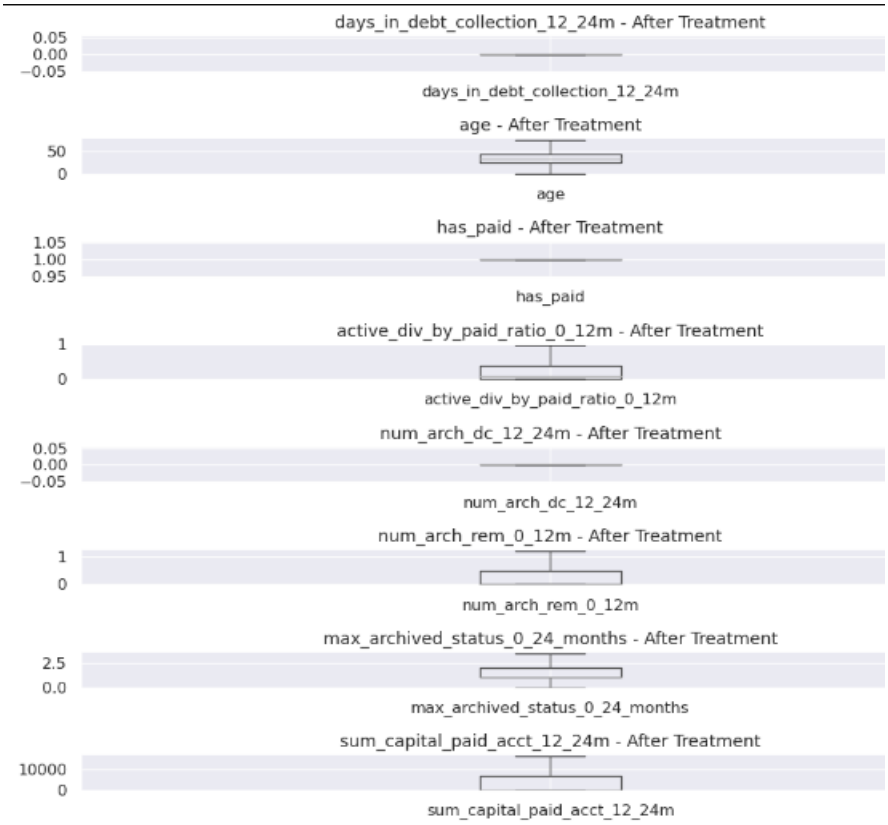

Figure 10: Boxplots after treating the outliers – 3

3f. Variable transformation (if applicable)

We have transformed the variable from ‘log_sum_paid_inv_0_12m’ to this 'sum_paid_inv_0_12m' by applying the log to the variable.

3g. Variable transformation (if applicable)

1.Total Purchases in the Last 12-24 Months: The line `df['total_purchases_12_24m'] = df['sum_paid_inv_0_12m'] + df['sum_paid_inv_0_12m']` calculates the total amount spent by the customer on purchases using the credit card in the last 12-24 months.

This variable is useful for understanding the overall spending behavior of customers over a specific period.

2. Active Divisions by Paid Ratio in the Last 0-12 Months: The line `df['active_div_by_paid_ratio_0_12m'] = df['num_active_invoices'] / df['sum_paid_inv_0_12m']` computes the ratio of the number of active invoices to the total amount paid by the customer in the last 0-12 months.

This variable provides insights into how actively customers are making payments relative to the number of invoices they have.

3. Maximum Archived Status in the Last 0-6 Months: The line `df['max_archived_status_0_6_months'] = df[['num_arch_dc_0_12m', 'num_arch_dc_12_24m', 'num_arch_ok_0_12m', 'num_arch_ok_12_24m', 'num_arch_rem_0_12m']].max(axis=1)` determines the maximum archived status among different categories (debt collection, OK, and reminder) in the last 0-6 months.

This variable helps identify the most severe status a customer account has been in over a specific period, providing insights into their payment behavior.

4. Maximum Archived Status in the Last 0-12 Months and 0-24 Months: The lines `df['max_archived_status_0_12_months'] = df[['max_archived_status_0_6_months', 'max_archived_status_0_6_months']].max(axis=1)` and `df['max_archived_status_0_24_months'] = df[['max_archived_status_0_12_months', 'max_archived_status_0_12_months']].max(axis=1)` extend the calculation of maximum archived status to the last 0-12 months and 0-24 months, respectively.

These variables provide a broader perspective on the historical account status and help in assessing customer risk over longer periods.

4.Business insights from EDA

4a. Is the data unbalanced? If so, what can be done? Please explain in the context of the business.

Sol: In the context of the business problem, an unbalanced dataset can lead to biased model performance, where the model may become overly optimistic about the default rate and may not effectively capture the patterns of default cases.

To address the imbalance in the dataset, several techniques can be employed:

1. **Resampling Techniques:**

Oversampling: Increase the number of instances in the minority class (default cases) by randomly duplicating them or generating synthetic samples using techniques like SMOTE (Synthetic Minority Over-sampling Technique).

Under sampling: Decrease the number of instances in the majority class (non-default cases) by randomly removing samples. However, this might lead to loss of information.

2. **Algorithmic Approaches:** Use algorithms that are inherently robust to class imbalance, such as ensemble methods like Random Forest, Gradient Boosting, or algorithms that allow specifying class weights, like Support Vector Machines (SVM) or Logistic Regression.
3. **Evaluation Metrics:** Choose appropriate evaluation metrics that account for class imbalance, such as F1-score, precision-recall curves, or area under the precision-recall curve (AUC-PR).
4. **Data Collection:** Collect more data, especially instances of default cases, to balance the dataset naturally.

4b. Any business insights using clustering (if applicable)

Sol: Clustering techniques can be applied to identify distinct customer segments based on their credit card usage patterns, purchase behavior, payment history, and other relevant features. These segments can provide valuable insights for targeted marketing strategies, risk assessment, and personalized customer services.

For example: **Segmentation by Spending Behavior:** Clustering customers based on their total purchases, average payment span, and merchant categories can reveal high-spending, low-spending, and moderate-spending segments. This information can be used to design tailored marketing campaigns and loyalty programs.

Risk Assessment: Clustering customers based on their payment history, days in debt collection, and maximum archived status can help identify high-risk, medium-risk, and low-risk segments. This segmentation can guide credit limit adjustments, targeted risk mitigation strategies, and proactive customer support initiatives.

4c. Any other business insights

Age-Based Analysis: Analyzing default rates across different age groups can provide insights into the creditworthiness of different customer segments. For instance, younger customers may have higher default rates due to limited financial stability, while older customers may exhibit more stable payment behaviours.

Merchant Category Analysis: Understanding default patterns across different merchant categories can help in assessing the risk associated with specific types of purchases. For example, customers making purchases in high-risk categories like entertainment or luxury goods may have higher default rates compared to those making purchases in essential categories like groceries or utilities.

Time-Based Analysis: Examining default rates over time can reveal seasonal trends, economic fluctuations, or changes in customer behaviours. This analysis can inform strategic decision-making and resource allocation, such as adjusting credit policies, enhancing collection efforts, or launching promotional offers during periods of higher default risk.

These insights can inform strategic decisions for the credit card company, including customer segmentation, risk assessment, and targeted marketing strategies, ultimately leading to improved customer satisfaction and business performance.